

Lukas Anell

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EDUCATION

- **Purdue University *GPA: 3.23*** West Lafayette, IN
BS in Data Science (CS) August 2025 - May 2029
- **South Pasadena High School *GPA: 3.7*** South Pasadena, CA
High School Diploma August 2021 - June 2025

EXPERIENCE

- **Let's Code** Monterey Park, CA
Coding Instructor July 2024 - July 2025
 - **Material:** Created project-based curricula to teach children ages 7-14 coding languages and critical thinking skills.
 - **Teaching:** Communicated effectively to manage classes of 25 students, collaborate with instructors, and train new interns.
 - **Feedback:** Assessed students' progress and provided feedback to enhance learning.
- **CS180 Team Project** Purdue University
Team Member (4-person team) Fall 2025
 - **Contribution:** Collaborated with a 4-person team to build a client-server movie reservation system in Java using socket-based communication. Implemented the **Reservation**, **Server**, and **User** classes and the Swing-based client GUI's booking flow.

PROJECTS

- **mini-database (April 2026 - Present) *Creator*:** Relational database engine built from scratch in Java supporting typed columns, CSV persistence, and SQL-like queries (SELECT/INSERT/DELETE/UPDATE with AND/OR/NOT WHERE clauses). Implemented dual indexing with a HashMap for O(1) equality lookups and a TreeMap for sorted range queries, and a snapshot-based transaction system supporting commit/rollback.
- **SnapTranslate (December 2025 - Present) *Creator*:** Chrome/Edge (Manifest V3) browser extension implementing a full capture → OCR → translate pipeline: drag-select any webpage region, extract text via Tesseract.js, and translate it through the DeepL API. Built a multi-context architecture (background service worker, content script, popup, offscreen document) with configurable OCR languages and per-user API key setup.
- **social-network (May 2026 - Present) *Creator* :** Social network engine modeling users as a weighted undirected graph with a custom persistence format. Implemented Dijkstra's algorithm for shortest paths, BFS/DFS traversal, mutual-friend-based friend suggestions, and community detection via connected components and label propagation.
- **HTTP/1.1 Web Server (May 2026 - Present) *Creator* :** Multithreaded HTTP/1.1 server built from raw sockets in Java with manual HTTP message parsing/construction. Implemented request routing (distinguishing 404 vs. 405), MIME-type-aware static file serving, and concurrent connection handling via a thread pool. In active development.

SKILLS

- **Languages:** Java, Python, Kotlin, C++, HTML/CSS, JavaScript/TypeScript
- **Software Engineering:** Data Structures & Algorithms, OOP, Git, Debugging, REST/HTTP, Socket Programming, Chrome Extension APIs
- **Data Science:** Python (pandas, NumPy, Matplotlib, scikit-learn, seaborn), R, SQL, Jupyter